

IT Helpdesk & IAM Home Lab — Project Documentation

Author: Mohd. Ismayeel Khan **Date:** 24 May 2026 **Tools Used:** VMware Workstation 17 Player, Windows Server 2022, Active Directory Domain Services

Introduction

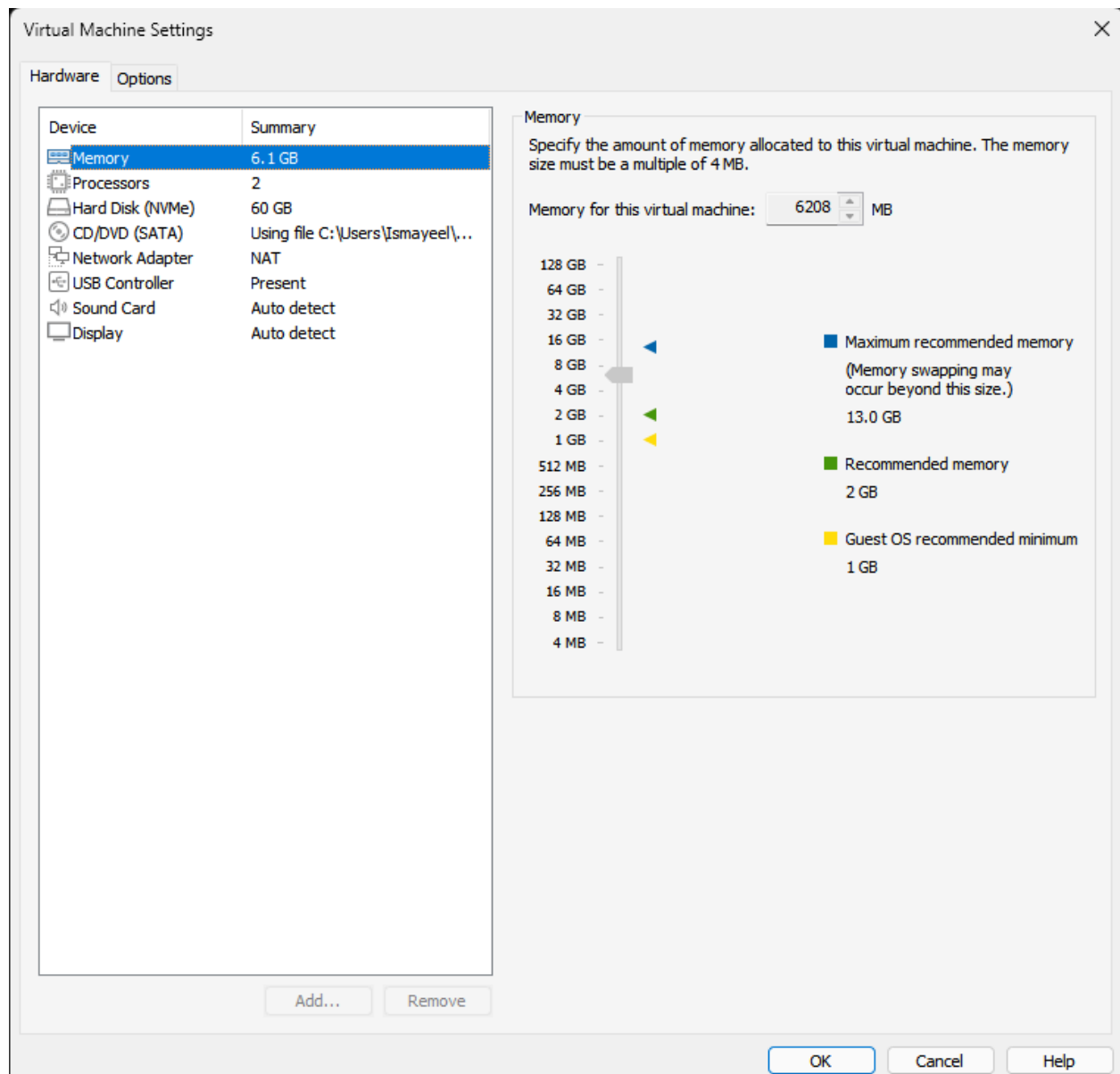
This project is a hands-on home lab I built from scratch to simulate a real enterprise IT environment. The goal was to practice and demonstrate skills relevant to both Helpdesk and Identity & Access Management (IAM) roles. Everything in this lab was done on my personal PC using free tools and evaluation software — no cloud, no shortcuts, just a real working environment built step by step.

Checkpoint 1 — Lab Environment Setup

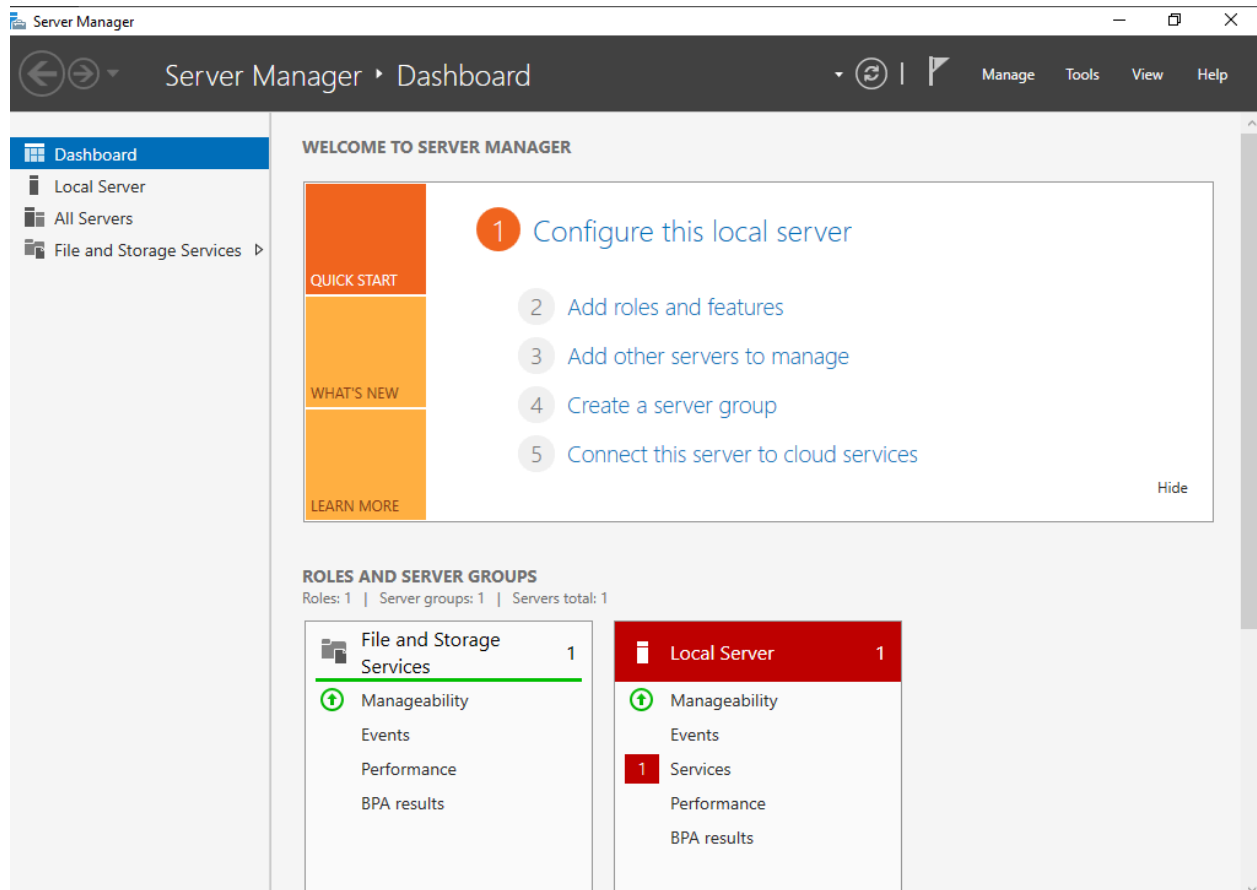
The first step was building the foundation. I downloaded the Windows Server 2022 Standard Evaluation ISO directly from the Microsoft Evaluation Center. This is the same server operating system used in real enterprise environments, which made this feel like working with production-level infrastructure from day one.

I then created and configured a Virtual Machine in VMware Workstation 17 Player with the following hardware specifications:

- RAM: 6GB
- CPU: 2 cores
- Disk: 60GB



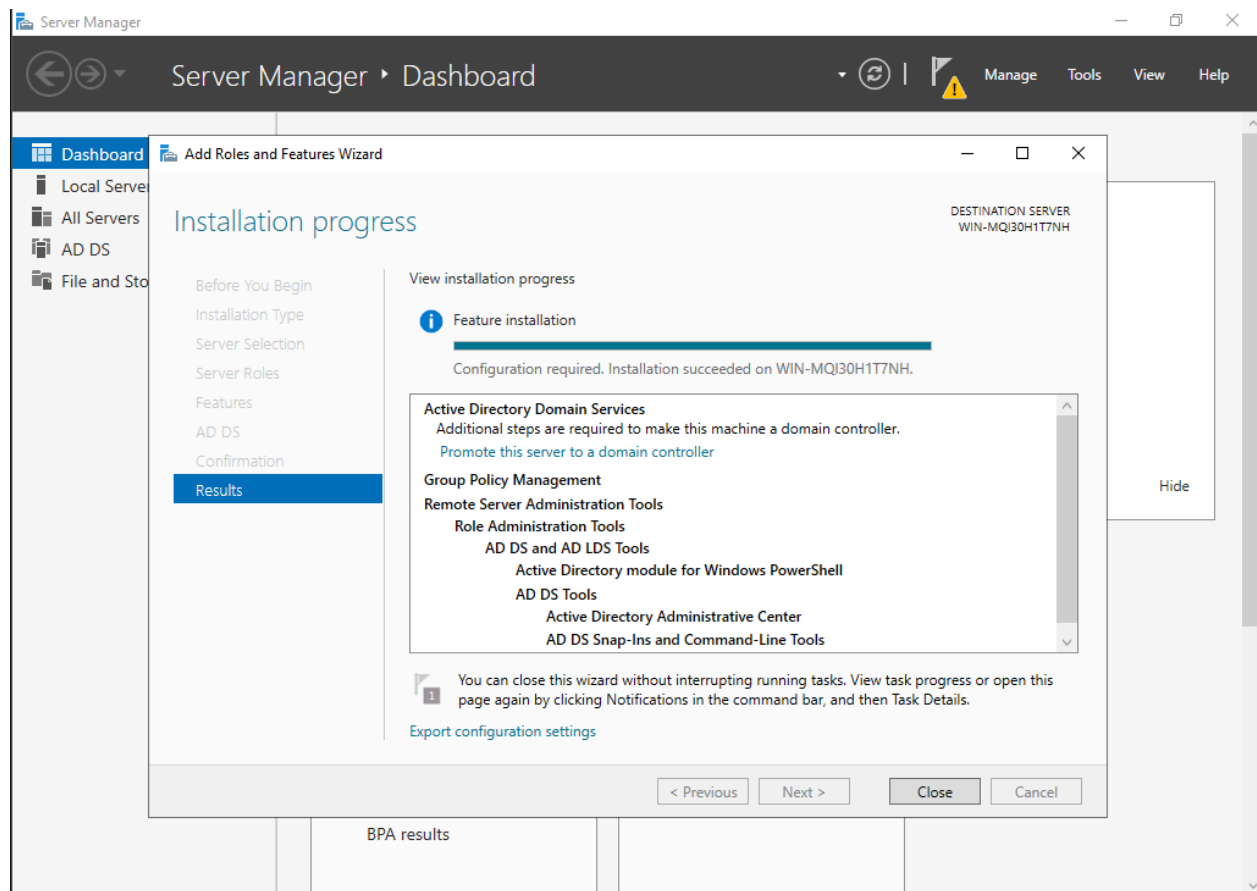
I selected Windows Server 2022 with Desktop Experience during installation to get the full graphical interface, making it easier to work with Server Manager and Active Directory tools. After installation I installed VMware Tools to improve VM performance and display integration.



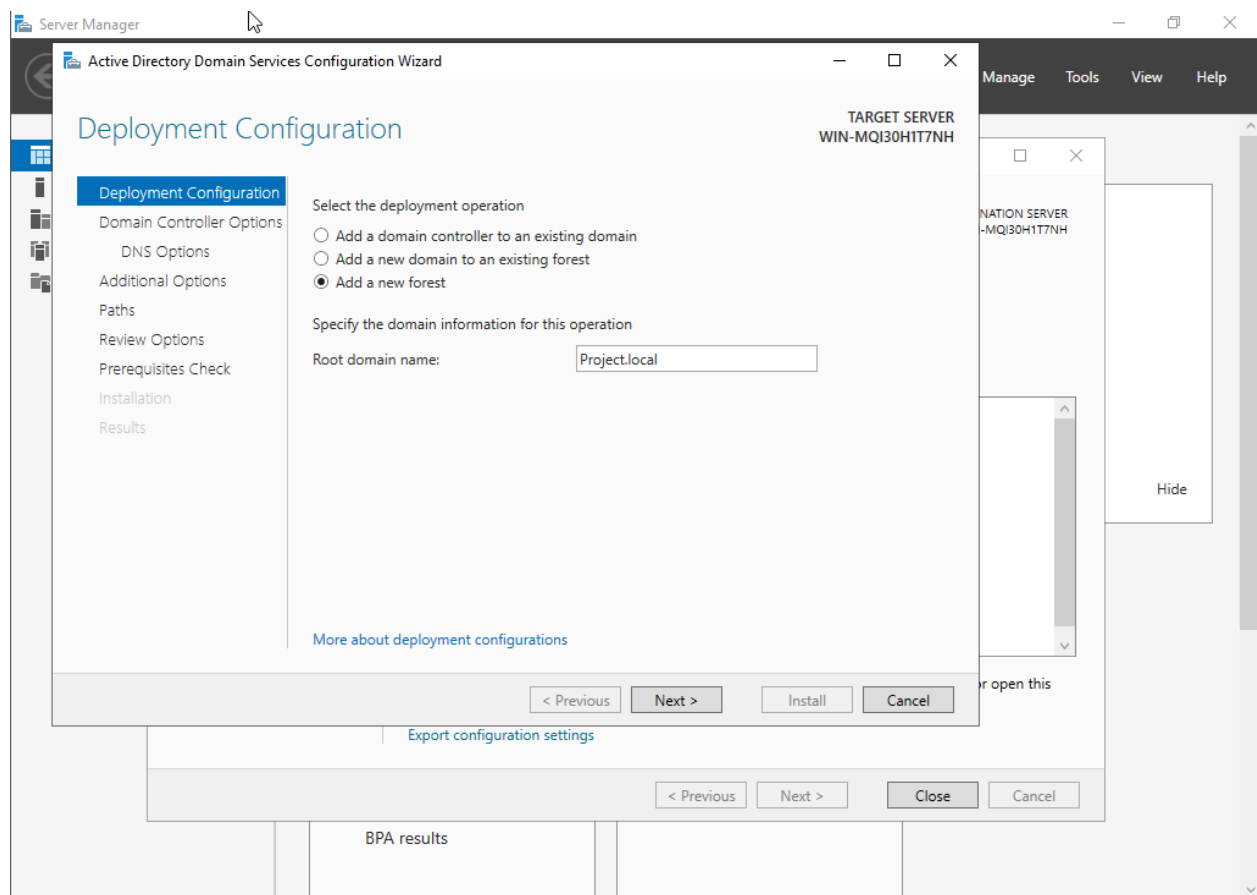
With the server up and running, the lab environment was ready to build on.

Checkpoint 2 — Active Directory Core

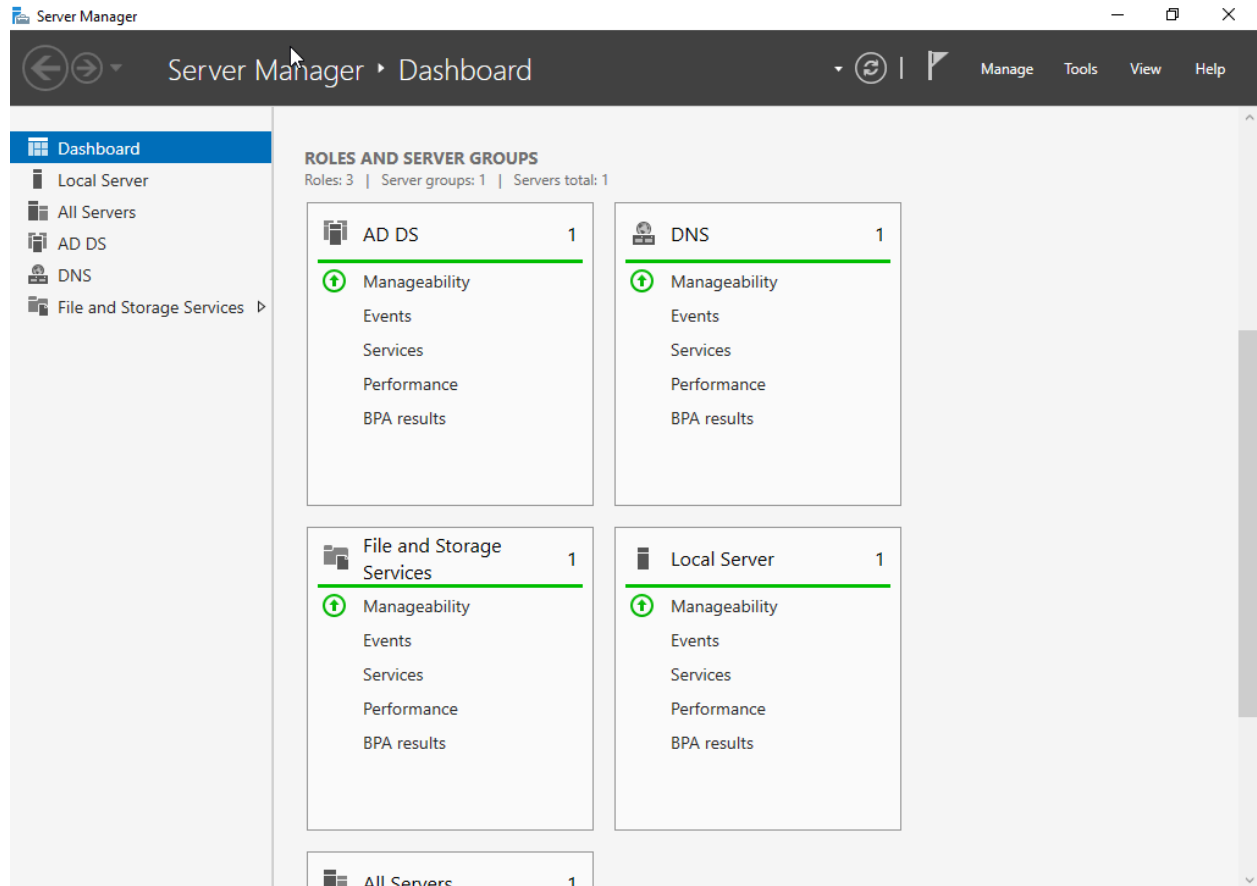
With the server running, the next step was turning it into a proper domain controller. I installed the Active Directory Domain Services (AD DS) role through Server Manager using the Add Roles and Features wizard.



After installation I promoted the server to a Domain Controller and created a new forest with the domain name **Project.local**. This is the core of the entire lab — everything else we build sits on top of this domain.



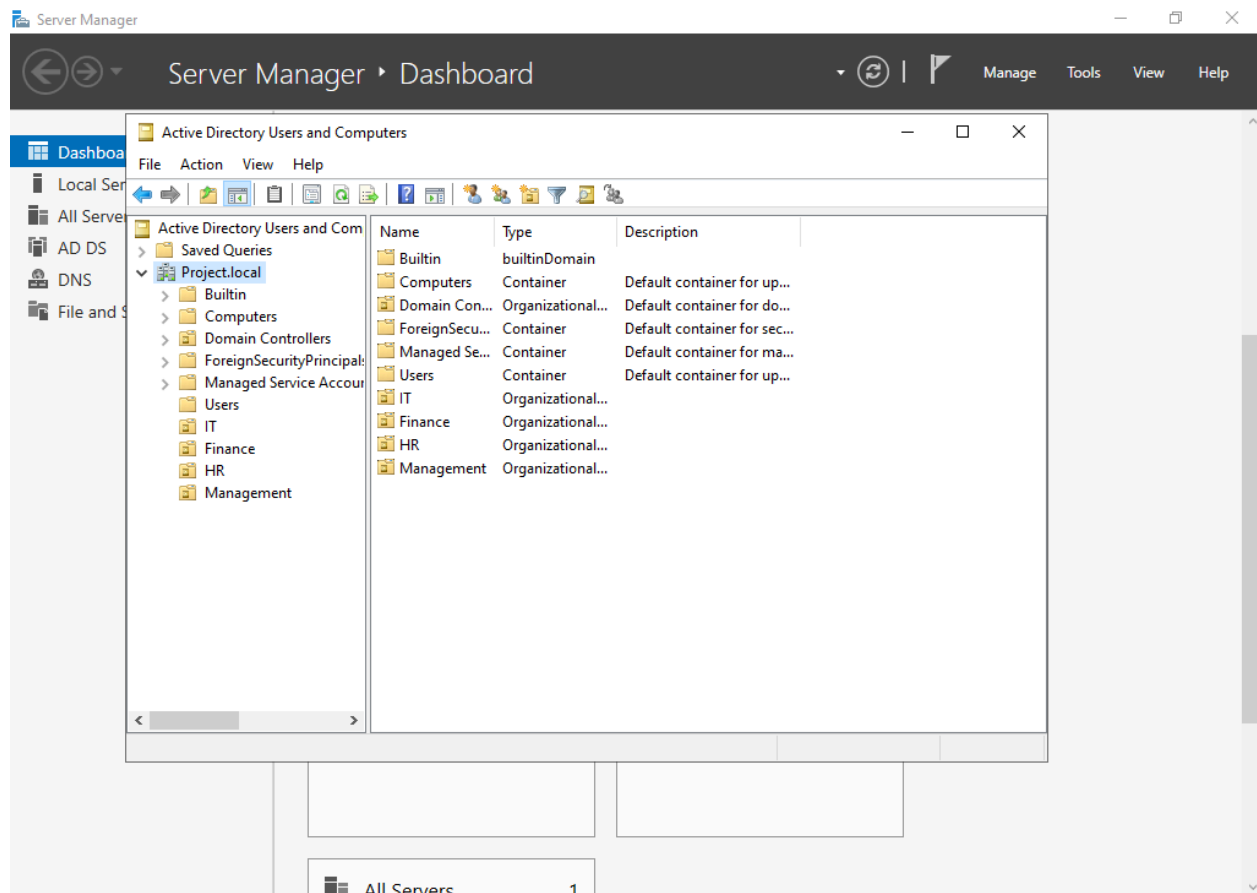
Once the server restarted and came back up as a Domain Controller, I could see both **AD DS** and **DNS** listed in the Server Manager Dashboard — confirming everything was working correctly.



Building the Organizational Structure

The next step was creating a realistic department structure using Organizational Units (OUs). I created 4 OUs to represent different departments in a typical company:

- IT
- HR
- Finance
- Management



Creating Users

With the OUs in place, I created a total of 15 users spread across all four departments to simulate a real workforce. Each user was created with a proper username format, a default password, and the "User must change password at next logon" option enabled — which is standard practice in enterprise environments.

Server Manager

Server Manager ▸ Dashboard

Manage Tools View Help

Active Directory Users and Computers

File Action View Help

Local Server

All Servers

AD DS

DNS

File and Storage Services

Active Directory Users and Computers

Saved Queries

Project.local

Builtin

Computers

Domain Controllers

ForeignSecurityPrincipals

Managed Service Accounts

Users

IT

Finance

HR

Management

Name	Type	Description
John Smith	User	
Sarah Conner	User	
Alena Costa	User	
Collette Torres	User	

All Servers 1

Server Manager

Server Manager ▶ Dashboard

ManageToolsViewHelp

Dashboard

Local Server

All Servers

AD DS

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ForeignSecurityPrincipals

Managed Service Accounts

Users

IT

Finance

HR

Management

Name	Type	Description
James Wilson	User	
Lisa Brown	User	
Alex Molina	User	
Haven Kelly	User	
Selena Garza	User	

All Servers

1

Server Manager

Server Manager Dashboard

Manage Tools View Help

Active Directory Users and Computers

File Action View Help

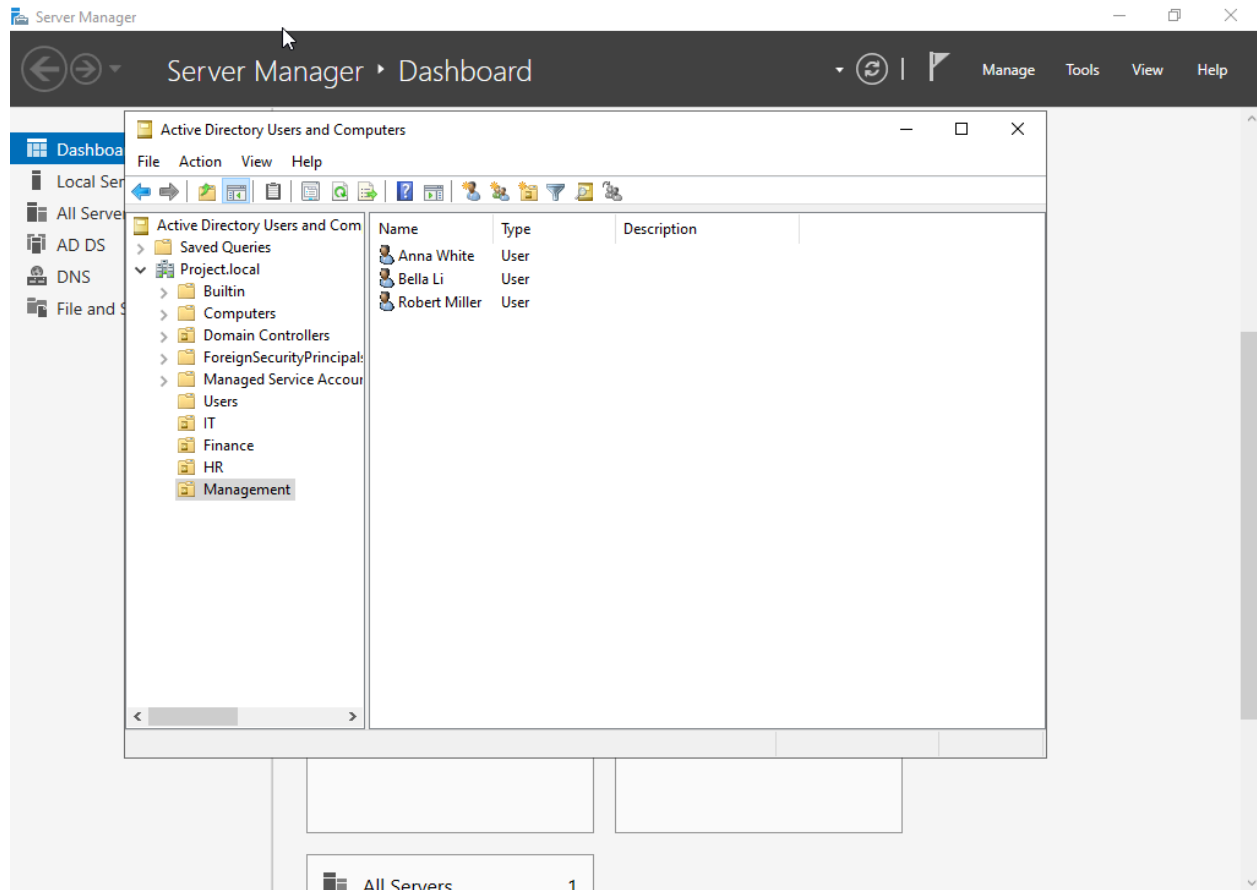
Local Server
All Servers
AD DS
DNS
File and Storage Services

Active Directory Users and Computers

- Saved Queries
- Project.local
 - Builtin
 - Computers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - Managed Service Accounts
 - Users
 - IT
 - Finance
 - HR
 - Management

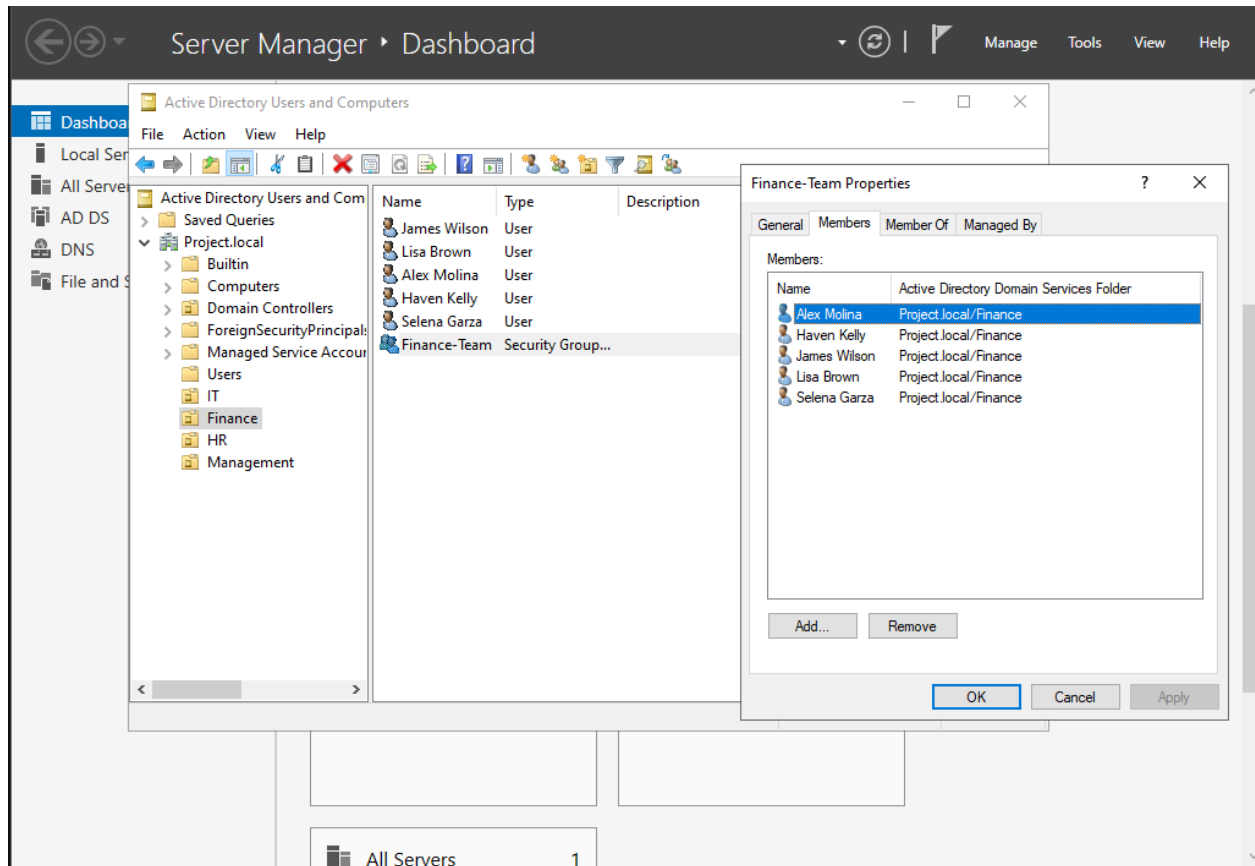
Name	Type	Description
Emily Davis	User	
Mark Taylor	User	
Erin Shelton	User	

All Servers 1



Security Groups

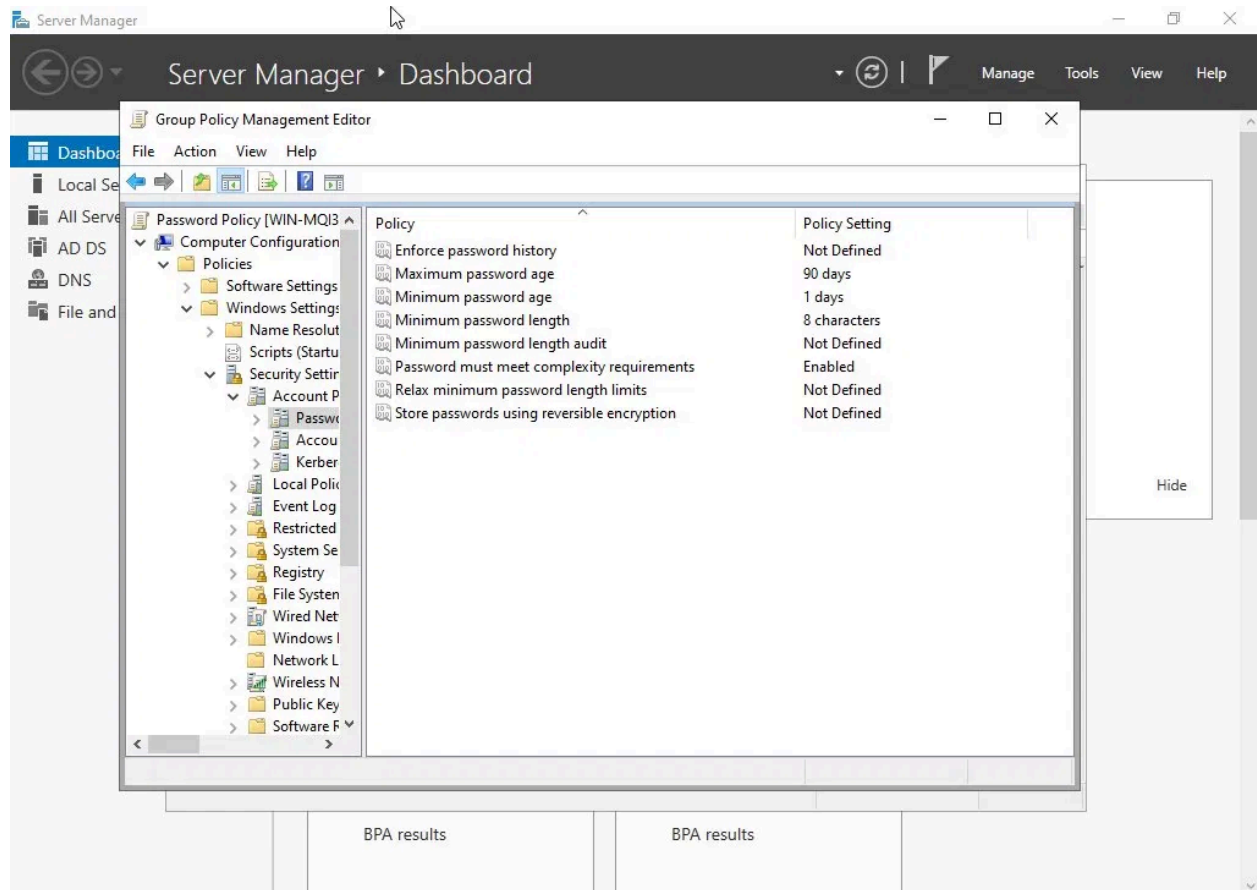
I then created a Security Group for each department — IT-Team, HR-Team, Finance-Team and Management-Team — and added the relevant users to their respective groups. Security Groups are a fundamental part of access management as they allow permissions to be assigned to a group rather than individual users, making administration much more efficient and scalable.



Password Policy — Group Policy Object (GPO)

Finally I created and linked a Password Policy GPO to the domain to enforce basic security standards across all users. The policy was configured with the following settings:

- Minimum password length: 8 characters
- Password must meet complexity requirements: Enabled
- Maximum password age: 90 days
- Minimum password age: 1 day



This kind of policy is standard in any organization that takes security seriously and is one of the first things an IAM analyst would review or configure when joining a company.